



Analysis

Insiders, outsiders, and the role of local enforcement in forest management: An example from Tanzania



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ABSTRACT

In low-income countries, both nearby local villagers, “insiders”, and non-locals, “outsiders”, extract products from protected forests even though their actions are illegal. Forest managers typically combine enforcement and livelihood projects offered to nearby communities to reduce this illegal activity, but with limited budgets cannot deter all extraction. We develop a game theoretic model of a forest manager’s decision interacting with the extraction decisions of insiders and outsiders. Our analysis suggests that, depending on the relative ecological damage caused by each group, budget-constrained forest managers may reduce total forest degradation by legalizing “insider” extraction in return for local villagers’ involvement in enforcement activities against outsiders.

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1. Introduction

During the last 30 years, the number of protected areas worldwide established to safeguard natural systems has grown dramatically. Coinciding with that expansion, many government agencies and conservation NGOs are advocating for combinations of development/livelihood policies and conservation/enforcement policies that attempt to address rural poverty and welfare while conserving forests (Naughton-Treves et al., 2005). For example, WWF’s policy on forests and poverty states that “national and international forest policies and the conservation movement should address both the sustainable management of natural forests and rural poverty alleviation; one should never be addressed at the other’s expense” (Gutman, 2001; p.9, para 1). The economics literature discussing policies aimed at conservation and poverty, such as Community-based Forest Management (CBFM), Joint Forest Management (JFM), and their predecessor Integrated Conservation-Development Projects (ICDPs), emphasizes their failure to create

incentives for conservation by rural people (see Hughes and Flintan, 2001, for a literature review; Behera and Engel, 2006; Ghimire, 1994; Johannesen, and Skonhott, 2005; Ligon and Narain, 1999; Muller and Albers, 2004; and Shyamsundar, 1996).

Even where there is an emphasis on poverty alleviation and the provision of livelihood projects for nearby villagers who lose access to forest resources, some level of enforcement of access rules is typically needed to deter illegal extraction, whether by locals or outsiders (Clarke et al., 1993; Robinson et al., 2010). Ostrom (1990) emphasized monitoring and enforcement as key aspects of protecting forests and forest resources (Chakraborty, 2001). Increasingly management policies such as CBFM require local people’s involvement in protecting the forests (Robinson and Lokina, 2012). An expanding literature finds that the involvement of local communities in monitoring and enforcing access and extraction rules tends to result in more favorable outcomes in terms of forest quality and reduced conflict. Gibson et al. (2005) highlights the debate over who should be responsible for enforcement, specifically whether governments should take on the role — as proposed by Bruner et al. (2001) and others; or local forest users — as emphasized by Stevens (1997) and Wells and Brandon (1992). Baland and Platteau (1996) stress the difficulties in protecting forests from outsiders particularly when markets develop for forest products such as

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wood fuels. More recently, Chhatre and Agrawal (2008) provide one of the first multi-country studies of the importance of local enforcement. The authors explore the relationship between forest regeneration and the extent of local enforcement using data that spans nine countries and find that the relationship between local enforcement and forest regeneration is complex but positive. That complexity depends in part on the extraction pressures on the particular forest, including whether people collect primarily for subsistence or commercial motives. Porter-Bolland et al. (2012) highlight the reality of many low-income countries: that most forests, even those under strict protection designations, are traditionally inhabited and managed by local people who extract various forest products. The authors suggest that in these situations forest cover is often better maintained than when these local people are excluded from the forest. Much of this literature emphasizes the actions of people located in or near protected forests. Yet many settings also face resource extraction by non-locals, which CBFM and related policies may not address adequately. For example, in West Bengal, India, forest protection committees were unable to restrict outsiders from collecting forest resources (Ravindranath and Sudha, 2004).

Our paper contributes to this expanding literature with a game-theoretic economic model of the interactions of forest manager and forest user decisions, while providing a new perspective on local-level community enforcement. The paper is motivated by Kibaha's forests in Tanzania, particularly the Ruvu North and South Forest Reserves, where forest managers struggle to protect the forests with limited budgets while facing illegal extraction pressure from both nearby villagers, "insiders", and non-local people, "outsiders." Insiders typically rely on the forests for fuelwood and other non-timber forest products, which is illegal given the forests' specific designation as a reserve. "Outsiders" illegally extract timber and produce charcoal, typically for sale in nearby Dar es Salaam.

In the following Section 2 we describe the particular situation in Kibaha district, Pwani Region. Motivated by a number of meetings with forest managers, patrollers, and rural villagers in Kibaha's Ruvu South forest reserve, we develop an economic modeling framework that incorporates the goals and tools of the forest reserve manager and the decisions of the two groups of resource extractors – insider village NTFP extractors and outsider charcoal producers – from those reserves, described in Section 3. Although meant to inform Kibaha's management directly, the model is sufficiently general to address a wide range of forest reserve settings with goals of forest protection and rural poverty alleviation. Section 4 concludes the paper with a discussion of the implications of our findings for including local people in forest management.

2. An Example from Tanzania: Kibaha's Forests

Despite the lack of well-established and documented mechanisms to induce conservation through poverty alleviation projects in and around parks, many parks still expect managers or NGOs to generate goodwill and achieve compliance with park regulations through a combination of enforcement activities and compensation for lost access to resources through poverty alleviation projects (Gutman, 2001; Wells and McShane, 2004). In Kibaha, we observed just such a combination of patrols and projects intended to provide the sticks and carrots needed to protect the nearby forests. Nearby villagers that we met with reported to us that they are pleased with the reserve management's tree planting in a designated buffer zone, and the provision of efficient stoves and beekeeping projects. But they are frustrated with charcoal producers from outside of local communities whose activities markedly degrade the forest (personal communication, village focus groups, Kibaha, 2011). Despite their concerns about widespread charcoal production, villagers tell us that they do not try to prevent outsider charcoal production nor contact forest guards about that production. After many discussions to understand why villagers do not report charcoal production, a group of women villagers described candidly their

own extraction behavior as it relates to the forest reserve-supplied beekeeping projects. When the women go to check on their beehives in the reserve forest, it is natural for them to collect forest resources, such as fuelwood and forest fruits and vegetables, at the same time. Although they often see outsiders making charcoal, they do not report them to the forest managers because the villagers' own collection is illegal. The villagers suggested that if their collection of forest resources was not illegal, they would have an unofficial mandate to report the charcoal producers to the forest manager, thereby improving the forest manager's ability to detect and punish charcoal production, given their small enforcement budget.

As in many settings, villagers have traditionally, though albeit illegally, relied on protected forests near their homes for important non-timber forest products (NTFPs). In addition, illegal extraction of logs and the in-reserve production of charcoal, often by non-local "outsiders," for sale in nearby Dar es Salaam cause serious forest degradation in addition to conflict with local people and forest guards (Godoy et al., 2012). Urban populations rely on charcoal for much of their energy needs and Kibaha's proximity to Dar es Salaam's large charcoal market makes illegal charcoal production a significant cause of degradation in Kibaha Forest Reserve (NBS, 2007; World Bank, 2009; interviews and personal communication, 2009 and 2010). We observed abandoned charcoal pits and young men leaving the reserve on bicycles laden with charcoal during our multiple visits to the reserve and surrounding villages. This charcoal extraction by and for outsiders reduces the ability of the forest to provide ecosystem services, whether provisioning, such as food and fuelwood collected by nearby villagers; regulating, such as climate control and carbon sequestration; or supporting, such as nutrient cycling (Millennium Ecosystem Assessment, 2005).

Many forests face different pressures from insiders and outsiders. For example, Alden Wily (1997) found in Tanzania's *miombo* woodlands that insiders used the forests for grazing, hunting, and charcoal production, while outsiders mainly were engaged in timber extraction. In Assam state, India, outsiders have been found mainly responsible for rhino poaching (Hussain, 2001). In Kibaha we found locals collecting a variety of non-timber forest products including fuelwood, medicines, vegetables, and thatching materials; while charcoal production (along with timber) is one of the most important forest products for outsiders (a similar finding to Haarstad et al., 2009).

The forest manager's typical tools to contain resource degradation include enforcement against extraction and income-generating livelihood projects, in addition to the less tangible goodwill and cooperation from nearby communities. Enforcement can be used to deter the illegal activities of both insiders and outsiders, whereas livelihood projects only provide incentives for households that live within project areas. Initiatives such as CBFM also often assume implicitly that goodwill will be generated through livelihood activities that will translate into "social fencing" by local villagers to protect the forest from illegal activity (Brandon and Wells, 1992; Lal, 1997; Lise, 2000; Nielsen, 2006). However, if local villagers collecting NTFPs such as fuelwood and forest vegetables face similar enforcement as outsiders producing charcoal, a potentially destructive forest-based activity, any goodwill generated through livelihood projects may well be lost. In Kibaha we found that forest managers hope that local villagers would be sufficiently invested in the forest reserve to help to enforce access restrictions against outsider charcoal producers, yet express dismay at the lack of such cooperation (personal communication in interviews, 2011). Motivated by informal discussions that we had with women living near to Kibaha's forests and findings from the literature, here we consider a third management tool for forest managers: defining some resource access rights for locals to motivate their cooperation in enforcing against outsiders.

3. Game Theoretic Model of Resource Extraction

We develop an economic model to explore some of the key issues raised by our observations in Kibaha. Our game theoretic modeling

structure considers explicitly the interaction between three actor groups for a homogenous forest. First, villagers living adjacent to the forest extract NTFPs such as fuelwood, which they value at local market prices because they can sell them in a nearby market or avoid purchasing those items. Second, charcoal producers, who come from outside the nearby villages, produce charcoal illegally in the forest and sell it in urban markets. Third, the forest manager (FM) has a limited fixed budget with which to protect the forest. The FM may additionally provide livelihood projects for local villagers such as bee keeping (in many instances of community-based management, these are funded from a different source, such as an NGO, see for example, Robinson et al., 2012); and may offer forest resource collection rights for these local villagers to encourage them to help enforce access restrictions against more damaging activities by outsiders. Villagers might be inclined to undertake this detection role due to the negative impact of charcoal production on NTFP and bee production or as a favorable response to the FM's granting of access rights. A key element of this model set up is the natural asymmetry between forest use by insiders and forest use by outsiders. Both groups extract from the forest, but they use the trees for different products and in different ways. The collection of fuelwood, forest fruits and vegetables, or forest medicines, as tends to be practiced by insiders, does relatively little damage to the forest. For example, villagers tend to collect dried dead wood or lop small branches for fuelwood. Though these activities can lead to degradation, they are rarely implicated in deforestation and the collection of these resources leaves individual trees intact. Bee keeping is a non-forest degrading activity that typically involves suspending a traditional beehive in a tree within the forest, or placing a non-traditional hive on the ground, also within the forest. In contrast, charcoal production typically involves the felling of whole trees, which thus reduces the ability of the forest to generate NTFPs such as fuelwood. Thus the resource, "trees", generates multiple benefits for different users, with different degradation implications resulting from harvest/production. Typically there are distinct markets for these different forest products: fuelwood and other NTFPs such as forest fruits and vegetables are predominantly used locally in the nearby rural areas; charcoal is predominantly for urban use. For example, over 70% of households in urban Dar es Salaam use charcoal (World Bank, 2009), but very few rural households. As such, although charcoal and fuelwood, a key NTFP, are substitutes for cooking, in practice few rural households use charcoal and few urban households use fuelwood. Thus one resource produces both products, but it is reasonable to assume these products are extracted/produced by different groups (insiders and outsiders) for different markets (rural and urban).

The access rights decision is akin to making a choice of the IUCN classification of a protected forest. We also compare the cases of with and without livelihood projects. As in Albers and Robinson (2011), we use beekeeping as our livelihood project because it is non-extractive, forest-dependent, and is a popular alternative income-generating project in Kibaha Forest Reserve and worldwide (see, for example, www.beesfordevelopment.org; Lietaer, 2009; Bradbear, 2004, 2009).¹ Indeed, Tanzania's Ministry of Natural Resources and Tourism (MNRT) manages forests through its Forestry and Beekeeping Division (MNRT, 1998, 2002a, 2002b, 2005). Through this framework we consider the tradeoffs between less restrictive protected area designations (access rights), levels of enforcement, and the introduction of livelihood projects.

3.1. Equilibrium with no Livelihood Project

We set up the model here for the case of no livelihood project. The three sets of players are the forest manager; n identical villagers; and a group of charcoal producers. This set up reflects the fact that many individual villagers tend to make independent decisions over how much

to collect from the forest, whereas charcoal tends to be produced by a group of individuals (one could envisage a model with multiple groups of charcoal producers, or one single villager, but such a set up would not produce additional insights). The forest manager's optimization is relatively simple (thus allowing us to focus on the insider/outsider interaction and the implication of extraction rights). He minimizes the ecological damage to the forest, recognizing that NTFP and charcoal production degrade the forest differently, subject to a fixed enforcement budget. His only choice is whether or not to legalize NTFP collection by local rural households.

A representative villager maximizes her expected returns, W_V , to labor spent collecting NTFPs, l_F , which may or may not be illegal depending on the FM's choice, subject to an opportunity cost of labor, k_F , that could reflect opportunities for wage labor or farming. Any NTFPs collected are sold in a nearby market.² Charcoal production is always an illegal activity. Charcoal producers maximize their expected returns W_C to labor, l_C , spent producing charcoal in the forest. The presence of charcoal production reduces the returns to villagers' NTFP collection effort.

Our model incorporates three key elements of rural resource extraction with incomplete enforcement: a "harvest" functions for NTFP extraction and charcoal production; and the enforcement regime.

3.2. Extraction Functions

With neither charcoal producers in the forest nor enforcement, following López-Feldman and Wilen (2008) and their paper addressing spatial aspects of forest resource extraction, we use a Schaefer-like function to represent total NTFP harvest, in which there are diminishing returns to total effort. We thus write total NTFP harvest (when there is no charcoal production) by the n villagers as:

$$H_F = R\alpha(nl_F)^\beta \quad [1]$$

R is the resource density in the forest before any extraction. The resource, which we can think of as the collection of trees that comprise the forest, can be used for both charcoal production and NTFP collection. Charcoal production involves chopping whole trees and has a tangible impact on the ability of the resource R to generate NTFPs. In contrast, NTFP collection is also a function of the resource density, R , but the collection of NTFPs does not affect charcoal production. This is an important but reasonable assumption for this paper, and drives the nature of the game theoretic interaction. α is a scaling parameter; and $\beta < 1$ such that there are diminishing returns to effort in the forest.

With charcoal production, H_C , by outsiders, the total village NTFP harvest, H_F , can be written:

$$H_F = R\alpha(nl_F)^\beta \left(1 - \frac{\gamma H_C}{R}\right) \quad [2]$$

γ is a scaling parameter, and $H_F + H_C \leq R$. Eq. (2) is a modified Schaefer function accommodating both "congestion" due to both multiple villagers and a degradation externality imposed on NTFP collection by a coordinated group of charcoal producers. The harvest of the individual villager, h_F , is therefore written:

$$h_F = \frac{H_F}{n} = \frac{R\alpha(nl_F)^\beta}{n} \left(1 - \frac{\gamma H_C}{R}\right). \quad [3]$$

With no enforcement, the charcoal harvest is $H_C = R\theta l_C^\phi$, and is not affected by the extent of NTFP collection. This modified Schaefer function recognizes diminishing returns to charcoal effort ($\phi < 1$).

¹ The projects discussed here do not include bark beekeeping but instead relate to creating beehives in a manner that does not degrade the forest as occurs with using locally lopped limbs or bark for the hives.

² We do not consider an NTFP resource requirement or minimum, as is often the case for fuelwood, but rather assume that there is a functioning and efficient market for NTFPs and villagers sell what they collect (in keeping with such papers as López-Feldman and Wilen, 2008).

3.3. Enforcement Regime

We introduce a simple enforcement regime (following the Becker, 1968, framework) in which some exogenous enforcement budget translates into a probability of detection q for both villagers (if NTFP collection is illegal) and charcoal producers. Without loss of generality, following Robinson et al. (2013), punishment is simply confiscation of the illegal NTFP/charcoal. This assumption fits with anecdotal evidence we have that it is rare for people to be taken to court and fined. Rather, confiscation of illegal extraction or abandonment of a charcoal oven is often the norm. Moreover, the key element of enforcement as a deterrent is the expected punishment, which is the probability of being detected multiplied by the cost of being detected and punished. Thus introducing a fine in addition to confiscation can be considered equivalent to increasing the probability of detection (though clearly the implications are different for the forest manager and the actual costs of implementing and enforcement policy).

If NTFP collection is illegal then the FM's enforcement efforts reduce the expected returns to both NTFP collection and charcoal production. If NTFP collection is legal, then villagers inform the FM of any illegal charcoal production that they witness, thereby supplementing the FM's enforcement effort. In this case, there is a strategic interaction between the three sets of players. We model that interaction as a Nash equilibrium in which the players move simultaneously. If NTFP collection is not legal then villagers do not contribute to enforcement against the charcoal producers, in keeping with our discussions with villagers in Kibaha, and there is no strategic interaction.

If villager collection of NTFPs is legal, village enforcement supplements FM enforcement and is proportionate to the time villagers spend in the forest collecting NTFPs. The probability of the charcoal producers being caught when villager enforcement supplements FM enforcement is:

$$q_F = q + ml_F, \quad q_F \leq 1, \quad \text{and} \quad m > 0 \quad [4]$$

where m is simply a scaling parameter. The more time villagers spend in the forest, the more likely that they will see and report illegal charcoal production.

3.4. Optimization

Individual villagers and the group of charcoal producers maximize their returns to labor. For the representative villager we let the opportunity cost of labor be k_F and for the group of charcoal producers k_C . Without loss of generality we let the price of a unit of forest products, whether extracted by villagers or outsiders, be one. The optimization functions accommodate both enforcement and congestion. If NTFP collection is illegal (local villagers are not involved in enforcement) the villagers' actions do not affect the charcoal producers' optimal labor allocation.

For the representative villager and charcoal producer, respectively:

$$\begin{aligned} \max_{l_F} [W_F] &= \frac{R}{n} \alpha (nl_F)^\beta (1 - \gamma \theta l_C^\phi) (1 - q) - k_F l_F \\ \max_{l_C} [W_C] &= (1 - q) R \theta l_C^\phi - k_C l_C. \end{aligned} \quad [5]$$

For the charcoal producers:

$$W_C' = (1 - q) \phi R \theta l_C^{\phi-1} - k_C. \quad [6]$$

For the representative villager:

$$W_F' = \frac{R}{n} \alpha \beta n^\beta l_F^{\beta-1} (1 - \gamma \theta l_C^\phi) (1 - q) - k_F. \quad [7]$$

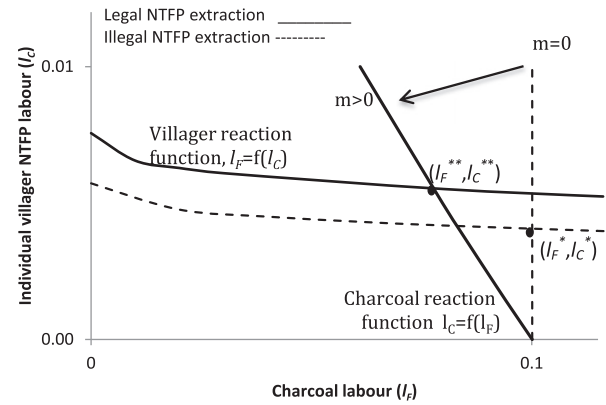


Fig. 1. Villager and charcoal producer reaction functions and equilibria as a function of villager access rules.

For an interior solution:

$$l_F = \left[\frac{k_F}{R \alpha \beta (1 - \gamma \theta l_C^\phi) (1 - q) n^{\beta-1}} \right]^{\frac{1}{\beta-1}} \quad \text{where} \quad l_C = \left[\frac{k_C}{(1 - q) \phi R \theta} \right]^{\frac{1}{\phi-1}}. \quad [8]$$

If NTFP collection is legal, the charcoal producers' optimization becomes a function of the insiders' decisions and the optimization equations become:

For the representative villager:

$$\max_{l_F} [W_F] = \frac{R}{n} \alpha (nl_F)^\beta (1 - \gamma \theta l_C^\phi) - k_F l_F. \quad [9]$$

For the charcoal producers:

$$\max_{l_C} [W_C] = (1 - (q + ml_F)) R \theta l_C^\phi - k_C l_C. \quad [10]$$

For this case of strategic interaction, again assuming an interior solution, solving the first order equations simultaneously results in the following reaction functions that define the Nash Equilibrium:

$$l_F = \left[\frac{k_F}{R \alpha \beta n^{\beta-1} (1 - \gamma \theta l_C^\phi)} \right]^{\frac{1}{\beta-1}} \quad \text{and} \quad l_C = \left[\frac{k_C}{(1 - (q + ml_F)) \phi R \theta} \right]^{\frac{1}{\phi-1}} \quad [11]$$

Fig. 1 shows the reaction functions for legal and illegal NTFP collection and the Nash equilibrium levels of labor allocated to charcoal and NTFPs under the different regimes.³ When NTFP collection is illegal and villagers are not involved in enforcement, the charcoal producer's reaction function is vertical because the villagers' actions do not affect the charcoal producers' production decision (dashed lines in Fig. 1). In this case, $m = 0$ and the equilibrium levels of labor allocated to NTFP and charcoal collection are given by (l_C^*, l_F^*) . When NTFP collection is legal, the charcoal producers' reaction function rotates counter-clockwise and demonstrates strategic interaction between the insiders and outsiders. The equilibrium levels of labor allocated to NTFP and charcoal collection are now (l_C^{**}, l_F^{**}) . Higher effectiveness of villager enforcement (proxied by m) leads to a more sloped charcoal producer reaction function.

Examining Fig. 1 reveals that the forest manager faces a trade-off when deciding whether or not to allow legal NTFP collection in exchange for villagers contributing to enforcement against charcoal producers. As compared to the illegal NTFP extraction case, with legal

³ To draw the graphs we use the following parameter values: $R = 20$; $\alpha = 0.6$; $\beta = 0.2$; $\gamma = 0.9$; $n = 100$; $k_F = 3$; $p_F = 1$; $\theta = 0.6$; $\phi = 0.35$; $\varphi = 0.6$; $k_C = 15$; $q = 0.2$; and $m = 22$. These parameters generate patterns and tradeoffs observed in Kibaha and other forests but do not reflect any particular setting.

NTFP collection, charcoal production is lower but NTFP collection is higher due to the lack of enforcement's deterrence (as represented by the time each player spends undertaking the forest-based activity). In general, which regime is preferred by the FM depends on his/her objective function, which conceptually could take into account how charcoal production and NTFP extraction affect the ecosystem services provided by the forest and/or villager or charcoal producer welfare. We focus on the impact of extraction only on the ecosystem services provided by the forest – the forest quality. Thus, the legalization of NTFP extraction poses a tradeoff between decreased charcoal production and increased NTFP production. From Fig. 1 we can see that if the forest manager does not consider villager welfare, whether the forest manager chooses to make villager NTFP collection legal depends on the relative ecosystem costs imposed by the villagers' and charcoal producers' extraction. Conceptually, we can draw "eco-isoquants" on Fig. 1 that show combinations of charcoal production and NTFP extraction for which the FM is indifferent in terms of the resulting ecological damage (in Fig. 2). Because the eco-isoquants represent a level of ecological damage, the FM prefers the Nash equilibrium located on the eco-isoquant closest to (0,0) where the damage is lowest. A more flat isoquant implies that the ecosystem damage caused by the collection of NTFPs is relatively high per unit of labor compared with the damage caused by charcoal production (Fig. 2a). Under such circumstances, the FM prefers to keep NTFP collection illegal, which implies no engagement of local villagers in enforcement against outsiders. More steep eco-isoquants imply that NTFP collection causes relatively little ecosystem damage compared with charcoal production. (Fig. 2b). Although our parameters are illustrative only, FMs in Kibaha describe this relationship between the two sources of ecosystem damage. In such settings, the FM prefers to legalize NTFP collection because the reduction in charcoal production through the enforcement effort of villagers more than offsets the additional ecosystem damage caused by the increase in NTFP collection. Although a unit of labor spent on NTFP collection might commonly disrupt the ecosystem less than that same labor in charcoal production, which eco-isoquant slope scenario applies in a given setting will depend on the sensitivity of the particular ecosystem and resource to that extraction. More generally, if the eco-isoquants are steeper than the gradient of a line joining the Nash Equilibria for legal and illegal extraction, then NTFP extraction will be legalized. If the eco-isoquants are less steep, then NTFP extraction will be kept illegal.

To summarize the implications of our model: if the FM is concerned only with the ecological damage caused by NTFP extraction and charcoal production, whether or not it is better to legalize NTFP collection in the forest depends on the effectiveness of villager enforcement effort (m) and the steepness of the eco-isoquants. With legal NTFP collection, villagers spend relatively more time in the forest, which increases the quantity of NTFPs collected for two reasons. First, more time spent collecting leads directly to more total extraction. Second, because the villager-augmented enforcement level leads to lower charcoal production, the resulting higher resource density and lower charcoal-induced congestion implies greater effectiveness per unit of labor in NTFP extraction. Therefore, not surprisingly, villagers unambiguously collect more NTFPs and are better off when NTFP collection is legal. The FM faces a trade-off between the increased resource degradation from NTFP collection and the decreased degradation from charcoal production. Steeper eco-isoquants reflect ecological situations in which charcoal production is more damaging than NTFP extraction, leading to a preference for legalization of NTFP extraction because those access rights lead to an overall reduction in the ecological damage to the forest.

In Kibaha, and as is assumed in this paper, insiders extract NTFPs while outsiders produce charcoal in the forest. The FM can thus distinguish between groups based on their extraction activities, which produce different levels of ecological damage. If both groups produced charcoal, the eco-isoquants would be steeper. Giving charcoal production rights to insiders would increase enforcement against outsiders. However, that enforcement would have to deter sufficient charcoal

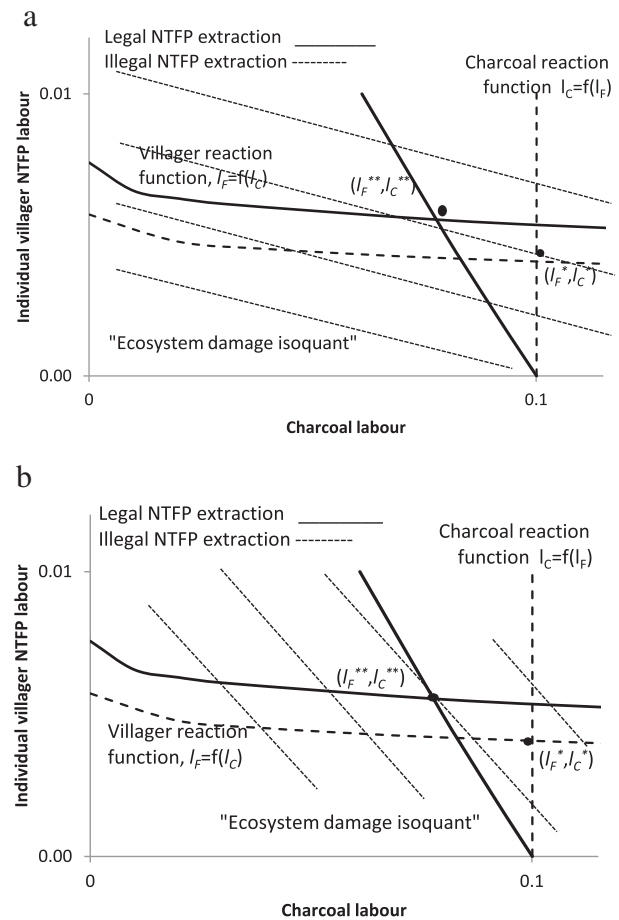


Fig. 2. 2a and 2b: Reaction functions and two possible eco-isoquants.

production by outsiders to offset the legal increase in charcoal production by insiders to make this an optimal strategy for an FM that focuses on minimizing ecological damage.

3.5. Impact of Introducing a Livelihood Project

Forest managers often consider offering villagers a livelihood project that may partially or fully compensate for lost livelihood benefits from reduced access to forest resources. In Kibaha, the forest managers have introduced beekeeping to the villagers. Beekeeping is a popular livelihood option from the forest manager's perspective because it is seen as creating an incentive for villagers both to protect the forest from charcoal producers and reduce their own forest degrading activities. Returns to beekeeping are higher in less-degraded forests, and villagers may reduce labor allocated to collecting forest resources when allocating labor time into beekeeping, thus leading to "conservation by distraction" (Ferraro and Simpson, 2006). The overall impact of the poverty alleviation project on the amount of ecosystem damage contains two dimensions working in opposite directions. First the project's value – returns to bee keeping – depends on the forest status, which depends on charcoal extraction. That dependence implies that villagers have a greater incentive to protect the forest against degradation in the bee-keeping case. Charcoal producers' reaction function rotates anti clockwise, implying a less steep reaction function, leading to a lower level of charcoal produced (in Fig. 2, imagine a less steep charcoal reaction function that pivots through the 0.1 point on the x axis). Depending on the location of the beehives, villagers could respond to the project by spending more time in the forest collecting NTFPs because they are also in the forest undertaking bee keeping activities. Working in the opposite direction, however, villagers could spend less

time in the forest collecting NTFPs if bee keeping were indeed an effective “conservation by distraction” policy. In that case, the villagers’ enforcement could decline overall, which would lead to a steeper charcoal reaction function and more charcoal produced. Thus it is not clear whether the total time spent in the forest, and so the total time for insider enforcement, would be greater or less than the no bee keeping option.

4. Discussion

Our paper is particularly relevant to situations where forest managers face different types of extraction by “insiders” and “outsiders”. To date, little academic or policy analysis provides information for managers in such settings about the best combinations of enforcement, poverty alleviation projects, and access rights. Our paper therefore makes a timely contribution, particularly as, increasingly, nearby villagers are expected to engage in protecting forests from which they have traditionally collected, often as an element of some type of participatory forest management regime. “Insider” villagers use nearby forests for extractive purposes and may have done so for many years (albeit often illegally) before the enforcement of restrictive forest management regimes; live adjacent to or close to the forest; and interact with forest managers through reserve-community projects or management. Typically, “outsiders” use the forest to collect marketed resources such as charcoal or timber; do not live locally; and have no rights or responsibilities in an active forest management regime. Although many forest managers create livelihood projects whose benefits accrue only to insiders, access rights and enforcement activities often treat insiders and outsiders equally despite potential differences between the groups in terms of their reaction to policy and the impact of extraction activities on the forest ecosystem.

Forest classifications with respect to access and extraction rules are often based on seemingly rational criteria, such as how vulnerable the particular ecosystem is to extraction, or how rare a particular species it contains. Yet the reality for many low-income countries is that there is insufficient budget to enforce strict access restrictions fully, and in such a second best situation, policies that attempt to exclude all extraction may not do as well as more nuanced policies. Robinson et al. (2013) suggests that spatial zoning, that could include a buffer zone within the protected area, can provide better overall protection than aspiring to protect the full area of a forest or park. This paper offers a complementary, but related, perspective: that a forest manager can get closer to their objective of reducing forest degradation by legalizing some activity that damages the ecosystem if, in return, there is less of a more damaging activity. In this paper, the forest manager trades off increased NTFP collection from nearby villagers in exchange for villager enforcement effort leading to less charcoal production by outsiders.

Though we demonstrate potential benefits of differential enforcement for insiders and outsiders through a game-theoretic model, in practice it can be difficult for a forest manager to distinguish between “insiders” and “outsiders”. In Kibaha, as is common where some type of participatory forest resource management is being implemented, “insider” villagers are clearly defined as those residing in specific villages that are typically adjacent to the forest under some type of PFM. Yet in practice identifying these individuals, and even who resides in a particular village is tricky. A similar issue has been documented in Tanzania’s marine protected areas (Robinson et al., 2012). In Mnazi Bay Ruvuma Estuary Marine Park, fisher people living inside the marine protected area (MPA), which explicitly enclosed 11 villages when it was set up, should have differential fishing rights. Yet the marine park managers told us that the reality is that they cannot tell whether fishers come from villages located inside or outside the MPA: insider fishers may have relatives living outside the MPA; or fishers might borrow insiders’ boats. Thus the park managers rely on enforcing fishing technologies rather than who fishes in

the MPA. Similar discussions have been had with respect to bushmeat hunting in west Africa (Robinson, 2008). Theoretically, legalizing hunting of common species could raise revenue from the sale of hunting permits that could be used to enforce against hunting of endangered species. But such selective hunting could be more tricky to enforce than an outright ban.

Even if such differential enforcement is tricky, it is not always clear why there should be insider villagers, or fishers, and why those individuals should have preferential rights over outsiders. In some cases, such as we find in many instances in Tanzania, the forest in question is situated within the boundaries of a particular village; but the terrestrial boundaries of a marine protected area may seem more arbitrary. Moreover, where forests provide global public goods such as carbon sequestration, who should benefit and why (Luttrell et al., 2012; 2013) remains a highly debated question (Louman et al., 2011).

Our paper assumes that the FM values the total quantity of resource in the protected area and does not take into account the impact of the protected area on forest users. But clearly one could adapt the model to take account of local and/or non-local villager welfare and different ecosystem services. The central point of the analysis remains however: that providing insiders with differential access rights – rights to extract NTFPs in the forest while outsiders have no such rights – could increase insider enforcement enough to reduce total ecological damage by deterring some outsider charcoal production.

Although in Kibaha the damage caused by outsiders appears to dwarf that by the insiders, in general, the impact of specific resource extraction activities on ecosystem services production determines the appropriate collection of differential access rights and enforcement. In particular, legalizing NTFP extraction presents an ecological tradeoff between increased NTFP extraction and decreased charcoal production due to insider enforcement against outsiders. Whether that ecological tradeoff generates net benefits relies on the ecological sensitivity of the system and the degree of change in both types of extraction resulting from the policy. We consider those two factors in defining theoretical eco-isocurves but little information on either component exists to inform forest managers’ decisions. In Kibaha, few estimates of the total quantities of biomass taken from the forests as fuelwood (mainly used by insiders) and charcoal (mainly used by outsiders) exist, with a complete dearth of information about the relative ecological impact of such activities and the likely responsiveness of extraction to policy changes. In addition, changes in fuel use over time could influence the level of damages caused by both insiders and outsiders. For example, if insiders move toward using charcoal that they produce in the forest while outsiders move toward electricity or gas in the cities, the contribution of insiders to the ecosystem damage could increase over time, which would limit the usefulness of targeting outsiders and suggest the need for targeting charcoal itself.

Given the preponderance of forests facing degradation from local and non-local people, especially in peri-urban forests, the game-theoretic framework here addresses a critical forest management issue. Despite the lack of data for site-specific management decisions, this analysis identifies the general interactions between insiders, outsiders, and forest managers’ decisions. Those interactions combine with the ecological setting to determine the best collection of enforcement, differential access rights, and livelihood projects.

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